

First Value, Not Latest: How Language Models Fail at Sequential Key-Value Update Tracking

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Abstract

We study sequential update tracking in interleaved streams: given a randomly interleaved stream in which K keys each receive N updates, can a language model retrieve the current value of a target key while competing keys are interspersed throughout? Across 20 models spanning 14 open-weight checkpoints (270M–9B parameters) and six proprietary frontier models, we find a robust asymmetry: accuracy on current-value queries (CVQ) is systematically lower than on first-value queries (FVQ), and the asymmetry persists across model scale and family. This failure is format-conditioned: adding explicit structural cues to the same content substantially reduces or eliminates the gap, while plain interleaved streams fail consistently. Our results indicate that language models apply format-conditioned retrieval heuristics over update streams rather than maintaining internal state, with direct implications for how sequential-update tasks are designed and evaluated.

1 Introduction

Language models are now deployed in settings that require tracking which value is currently assigned to an entity as it changes over time: multi-turn assistants where users correct facts mid-conversation, retrieval-augmented pipelines where documents may reflect outdated state, and tool-augmented agents where successive API calls update the same variable. Laban et al. (2025) show that multi-turn dialogues cause substantial performance degradation as state updates cascade and contradict earlier context. These failures share a common structure: a model that anchors to an earlier value and fails to incorporate subsequent updates.

Prior work on entity state tracking (Kim and Schuster, 2023; Kim et al., 2024) tests models on single-entity sequential transitions — one object moved between locations — without competing entities simultaneously present in the stream.

Stream ($K=3$, $N=3$, randomly interleaved):

color: red size: large shape: round
size: small color: blue shape: square
shape: triangle color: green size: medium

FVQ: What was the *first* value of color? → **red**

CVQ: What is the *current* value of color? → **green**

Figure 1: Task illustration ($K=3$, $N=3$). The stream randomly interleaves updates for all K keys. To answer CVQ correctly, a model must isolate color entries from competing keys and identify the most recent one among them.

Wang and Sun (2025) demonstrate KV interference within context but frame it as a fundamental architectural limit. Critically, neither line of work examines the cross-key competition that characterises real multi-entity tracking: multiple entities whose updates are interleaved throughout the stream, each potentially masking the others. Liu et al. (2024) and long-context benchmarks such as RULER (Hsieh et al., 2024) and HELMET (Yen et al., 2024) test whether models can retrieve static facts from noise — not whether they can identify which update to a repeatedly-modified key is current.

We study this gap directly: given a randomly interleaved stream of K keys each updated N times, can a model retrieve the current value of a target key while competing keys are interspersed throughout (Figure 1)? Across 15 models spanning 270M to 9B parameters, we find a robust asymmetry between first-value and current-value accuracy across all tested models and scales, and show the failure is format-conditioned: explicit structural cues and narrative format substantially restore performance while plain streams fail consistently.

We make the following contributions:

- A systematic study of sequential update tracking across 20 open-weight and proprietary

models on a controlled $K \times N$ grid, establishing a robust FVQ/CVQ asymmetry across transformer families and scales; results on Arbitrary-Single confirming the pattern are in the appendix (§4).

- A format ablation showing the asymmetry is format-conditioned: explicit structural markers and narrative format substantially reduce or eliminate the gap, while plain interleaved streams fail consistently (§6).
- A positional analysis showing that models can track only the first few positions in plain update streams before accuracy collapses, indicating a structural enumeration deficit rather than a recency-specific limitation (§5).
- A mechanistic analysis using probing classifiers and logit lens, revealing that CVQ correctness is not reliably encoded in final-layer representations, and identifying two distinct internal failure modes: *first-value anchoring* — the first value dominates all layers from the outset — and *transient suppression* — the correct current value briefly surfaces at intermediate layers but is suppressed before output (§7).
- A training-dynamics analysis across 82 checkpoints from two model families — 42 pre-training checkpoints of SmoLM2-1.7B and 40 SmoLM3-3B checkpoints spanning three pretraining stages and four post-training stages (mid-training, SFT, expert specialisation, alignment soup) — showing the asymmetry is present at the earliest pretraining checkpoint of both models, persists through every subsequent training step, and survives every post-training stage including the final released model (§7, Appendix G).

2 Related Work

Sequential state tracking. Early work established entity state tracking as a hard problem for language models: Weston et al. (2015) used synthetic narratives and Dalvi et al. (2018) used procedural text. Kim and Schuster (2023) found that only GPT-3.5 could track entity state in programmatic transition sequences, attributing this to code pretraining (Kim et al., 2024). These works evaluate only final-state accuracy using action-verb

prose; none study the asymmetry between first-value and current-value retrieval, nor plain interleaved streams with multiple competing keys.

KV update tracking. Most directly related, Wang and Sun (2025) study a KV update setting where 46 keys each receive up to 400 updates, finding retrieval accuracy declines log-linearly as updates accumulate. They also test format interventions — natural language annotations marking prior values as outdated — and find minimal improvement, concluding the failure is fundamental. Our work differs in three ways: we test both first-value and current-value queries, revealing an asymmetry their single-query design cannot observe; we apply structural format interventions (positional labels and round markers) rather than semantic annotations, and find these substantially restore performance; and we vary key count K systematically, identifying it as the primary driver of failure.

Long-context retrieval. Liu et al. (2024) show that models exhibit a U-shaped accuracy curve in long contexts, reliably accessing information near the beginning and end but failing at mid-context positions. Hsieh et al. (2024) introduce a variable-tracking sub-task where a value V is assigned to X_1 , followed by a chain of bindings ($X_2=X_1$, $X_3=X_2$, ...) embedded in noisy filler text; each variable is assigned *once*, and the model must identify all variables pointing to V . Our task differs from both: we test repeated destructive updates to the *same* key interleaved with competing keys, and our failure scales with the number of competing keys rather than context length or chain depth.

Format sensitivity and theoretical limits. Sclar et al. (2024) show that meaning-preserving changes to prompt formatting cause accuracy swings of up to 76 percentage points; our format ablation is a controlled instance of this, holding stream content constant while varying only its positional structure. Fagnou et al. (2024) prove that tracking compositional entity chains — where each state depends on the prior — requires $\lceil \log_2(n+1) \rceil$ transformer layers; our task involves independent destructive updates with no dependency chain, so this depth constraint does not apply. Hahn (2020) shows that softmax attention dilutes the influence of any single token as sequence length grows, and Merrill and Sabharwal (2023) proves that fixed-depth transformers are bounded by first-order logic with majority quantifiers: in a plain interleaved stream where

all entries for a key are structurally identical, attention over those entries disperses rather than concentrating on the most recent one. Our format interventions are consistent with this: explicit positional labels give each entry a unique token signature, allowing attention to concentrate on the correct position and converting implicit sequential tracking into an explicit lookup. Li et al. (2025) apply linear probes and activation patching to models trained on permutation composition, identifying two interpretable tracking algorithms in models that *succeed* at the task. We apply the same toolkit to models that *fail* at destructive key-value update tracking—a direction prior work leaves unaddressed—and identify two distinct failure modes across model families.

3 Task and Experimental Setup

3.1 Task Definition

We define **Sequential Update Tracking under Interference** (SUTI): given an interleaved stream of key-value pairs in which K keys each receive N updates, retrieve the current (most-recently assigned) value of a designated target key.

Formally, let $\mathcal{K} = \{k_1, \dots, k_K\}$ be a set of keys and let $V_i = (v_{i,1}, \dots, v_{i,N})$ be the ordered update sequence for key k_i . The input stream σ is a randomly interleaved permutation of all $K \times N$ pairs $\{(k_i, v_{i,j})\}$, subject to the constraint that no two consecutive entries share the same key. The model receives σ and a target key k_t , and is asked one of two queries:

- **FVQ** (first-value query): “What was the first value of k_t ?” $\rightarrow$ expected answer $v_{t,1}$
- **CVQ** (current-value query): “What is the current value of k_t ?” $\rightarrow$ expected answer $v_{t,N}$

Figure 1 illustrates the structure for $K = 3$, $N = 3$. The stream contains $K \times N = 9$ entries; the correct CVQ answer for *color* is *green* (its third update). We call K the **key count** and N the **update depth**. The two dimensions are confounded with stream length ($K \times N$ total entries); we address this in Section 4.

3.2 Stream Formats

We evaluate three prompt formats applied to the same underlying stream content (Figure 2).

Plain stream. Key-value pairs are written as key: value, one per line, randomly interleaved. No positional markers are present. This is the primary format for our behavioral sweep.

Labeled stream. Each entry is annotated with an explicit update counter inline: key (update j): value. The model can identify any specific update by its label without enumerating the stream.

Block stream. All K keys are grouped together per round under a [Round j] header. Updates are not individually labeled but round boundaries are explicit.

We use these formats in Section 6 to test whether CVQ failure is format-conditioned. The plain stream is the failure condition; labeled and block streams test whether externalising positional structure restores performance.

3.3 Datasets

We evaluate on two stream formats: a structured **key-value list**, used for the main grid and mechanistic analysis, and a natural-language **narrative**, used for the format intervention in §6. The key-value variant uses 46 semantic categories (e.g., *color, fruit, metal*) following the protocol of Wang and Sun (2025); the narrative variant is a Dota 2 match commentary dataset in which heroes act as keys and attribute updates (gold, kills, ...) act as values woven into match prose. The mechanistic analysis in §7 uses a single-token variant of the key-value format to make activation patching tractable. Full construction details for both formats appear in Appendix A.

3.4 Experimental Grid

We evaluate all models on a unified grid:

$$K \in \{2, 3, 5, 7, 10, 15, 20, 25, 30\}, \quad N \in \{5, 7, 10, 15, 20, 30, 50\}$$

with up to 200 trials per (K, N) cell per query type (FVQ and CVQ). Cells excluded for context-length or pool-size reasons are detailed in Appendix A.

3.5 Models

We evaluate 14 open-weight models (270M–9B parameters) spanning Qwen2.5, Qwen3.5, Gemma-3, and three LLaMA-style models (TinyLlama-1.1B, StableLM-2-1.6B, Pythia-410M), together with six proprietary frontier models. For the figures and mechanistic analysis in the main text we focus on

Plain stream	Labeled stream	Block stream
color: red size: large color: blue size: small color: green size: medium CVQ: current color? Model: red ✗	color (update 1): red size (update 1): large color (update 2): blue size (update 2): small color (update 3): green size (update 3): medium CVQ: current color? Model: green ✓	[Round 1] color: red size: large [Round 2] color: blue size: small [Round 3] color: green size: medium CVQ: current color? Model: green ✓

Figure 2: The three stream formats used in our experiments ($K=2$, $N=3$). Same content, different structure. The **plain stream** provides no positional markers; the model must enumerate the stream internally to find the most recent entry. The **labeled stream** tags each entry with an explicit update index, making any position directly retrievable. The **block stream** groups all keys per round under a round header, making round boundaries explicit without per-entry labels. Typical model responses shown; **(NOTE: replace with actual model outputs from experiment data before submission.)**

two models from different families, **Qwen2.5-3B-Instruct** and **Gemma-3-4b-it**, chosen for their distinct failure shapes (§4). Cross-model breadth on Semantic-Multi appears in Table 1; the full per-model grid and Arbitrary-Single replication are deferred to Appendices D and E.

3.6 Evaluation

For each (K, N) cell, we report FVQ accuracy and CVQ accuracy as exact-match proportions, with Wilson score 95% confidence intervals (Wilson, 1927). The primary metric is the **gap**: $\Delta = \text{FVQ} - \text{CVQ}$. A positive gap indicates FVQ dominance (the failure mode we study); a negative gap indicates CVQ dominance. All inferences use greedy decoding (temperature 0) to eliminate sampling variance.

4 The FVQ–CVQ Asymmetry

Every model we evaluate is more accurate at recovering the *first* value bound to a key than the *current* value, on at least one (K, N) cell of our grid. Figure 3 shows the shape of the failure for two representative open-weight models, and Table 1 reports a representative cross-model summary on Semantic-Multi.

Shape of the failure. Figure 3 traces FVQ and CVQ accuracy along two axes of the grid for Qwen2.5-3B-Instruct and Gemma-3-4b-it. In Gemma-3-4b-it FVQ tracks ceiling while CVQ degrades monotonically in both K (number of competing keys) and N (updates per key) — the canonical asymmetric pattern. In Qwen2.5-3B-Instruct both FVQ and CVQ degrade, but CVQ degrades faster and stays below FVQ throughout the scanned

Model	FVQ	CVQ	Gap
<i>Proprietary models, $K=20$, $N=50$</i>			
GPT-4.1	1.00 ± 0.01	0.71 ± 0.07	+0.29
GPT-4.1-mini	1.00 ± 0.01	0.74 ± 0.06	+0.26
Claude-4.5-Haiku	1.00 ± 0.01	0.53 ± 0.07	+0.47
Claude-Sonnet	1.00 ± 0.01	0.74 ± 0.07	+0.26
Gemini-2.5-Flash	1.00 ± 0.01	0.54 ± 0.07	+0.46
Gemini-2.5-Pro	0.90 ± 0.05	0.77 ± 0.06	+0.13
<i>Open-weight models, $K=5$, $N=30$</i>			
Qwen2.5-3B-Instruct	0.61 ± 0.09	0.55 ± 0.10	+0.06
Qwen3.5-2B	0.62 ± 0.09	0.48 ± 0.10	+0.14
Qwen3.5-4B	0.86 ± 0.07	0.64 ± 0.09	+0.22
Qwen3.5-9B	0.86 ± 0.07	0.79 ± 0.08	+0.07
Gemma-3-1b-it	0.55 ± 0.10	0.02 ± 0.03	+0.53
Gemma-3-4b-it	0.97 ± 0.04	0.68 ± 0.09	+0.29

Table 1: FVQ and CVQ accuracy on Semantic-Multi. Up to 200 trials per cell; values are accuracy $\pm$ Wilson 95% half-width. Each block reports the smallest (K, N) cell at which every model in the block shows gap $> +0.05$.

region. The asymmetry is therefore not a single phenomenon: in some models it takes the form of a failure to overwrite a stable first binding, in others a faster loss of the most recent binding under load.

Cross-model breadth. Table 1 reports the asymmetry across twelve models on Semantic-Multi. Because the cell at which a model first exhibits a measurable gap depends on scale, we report each block at the smallest (K, N) where every model in that block shows gap $> +0.05$: $K=20$, $N=50$ for proprietary models, $K=5$, $N=30$ for open-weight models. Scale shifts the failure regime — proprietary models hold the ceiling on cells where open-weight models already collapse — but does not eliminate the asymmetry.

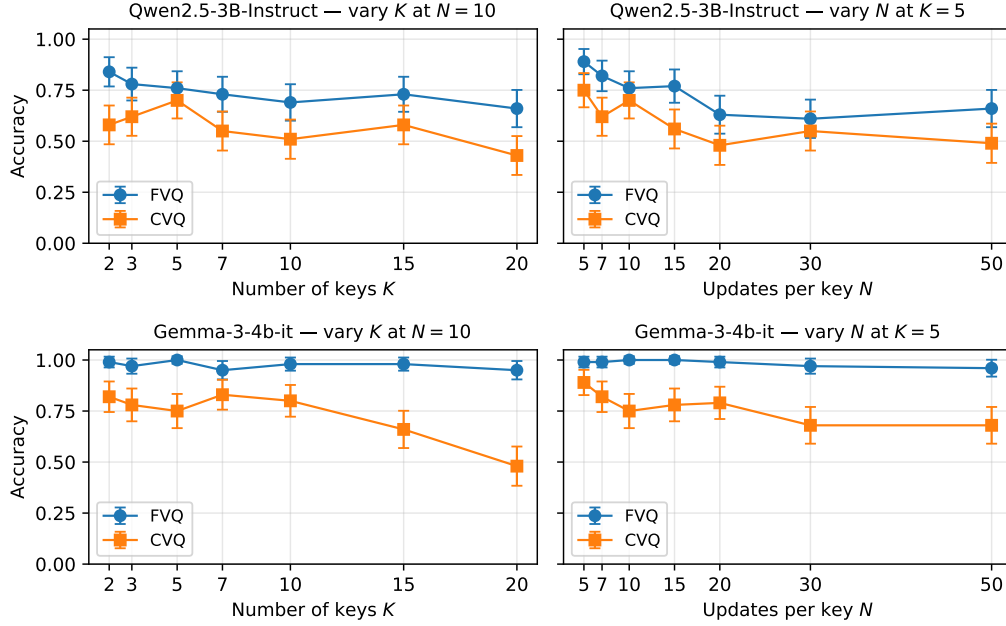


Figure 3: FVQ and CVQ accuracy on Semantic-Multi for two open-weight models. **Top row:** Qwen2.5-3B-Instruct. **Bottom row:** Gemma-3-4b-it. **Left:** vary number of keys K at fixed update depth $N=10$. **Right:** vary update depth N at fixed key count $K=5$. Error bars are Wilson 95% confidence intervals. CVQ sits below FVQ at every cell shown.

(NOTE: Qwen2.5-3B-Instruct reverses to CVQ > FVQ at $K \geq 10, N \geq 15$ on SEM (e.g., -0.27 at $K=10, N=50$); not used in the figure or table above. We defer this to §7, where it is related to the magnitude of the discriminating attention heads.)

The behavioural asymmetry shown above motivates the rest of the paper. §5 examines whether the failure is positional — which entries within a key’s update sequence the model can and cannot recover. §6 shows that structural format changes (positional labels, round blocks) substantially close the gap without changing the underlying content, indicating the failure is format-conditioned rather than informational. §7 identifies the internal mechanism: a specific cluster of attention heads in the upper-middle layers carries the FVQ–CVQ distinction in both Qwen2.5-3B-Instruct and Gemma-3-4b-it, and the asymmetric magnitudes of their primacy and recency boosts predict the behavioural asymmetry observed here. Full per-model (K, N) grids and results on the controlled Arbitrary-Single (ARB) dataset — which qualitatively replicates the SEM patterns — appear in Appendices D and E.

5 Positional Analysis

6 Format Interventions

7 Analysis

7.1 Training dynamics

To rule out the asymmetry being an artifact of alignment, supervised fine-tuning, or any single training step, we track the FVQ–CVQ gap across 82 checkpoints from two model families: 42 pretraining checkpoints of SmolLM2-1.7B (step 125k through the released final model; Ben Allal et al., 2025) and 40 SmolLM3-3B checkpoints spanning three pretraining stages and four post-training stages (mid-training, SFT, expert specialisation, alignment soup; Hugging Face Smol Models Research Team, 2025). At each checkpoint we evaluate on Arbitrary-Single at $K \in \{2, 3, 5, 7\}, N \in \{2, 3, 5, 7, 10, 15, 20, 30\}$ (100 trials/cell, completion format throughout to avoid chat-template artifacts at early IT checkpoints; see Appendix G).

Figure 4 reports the mean gap at each checkpoint, partitioned into two regimes that show qualitatively different trajectories. For **low-load cells** ($K \leq 3$; mean over 16 cells per checkpoint), the gap is positive at the earliest available checkpoint in both models and stays positive across the entire trajectory — the primacy regime. For **high-load cells**

($K \geq 5$ and $N \geq 20$; mean over 4 cells per checkpoint), the gap is *negative* at the earliest checkpoint in both models and stays negative across the entire trajectory — the recency regime, in which CVQ accuracy exceeds FVQ.

The reversal we noted in §4 for Qwen-family models at high (K, N) is therefore not an idiosyncrasy: the same regime flip is established within the first 40k–125k pretraining steps of both SmoLLM2 and SmoLLM3, persists through every pre-training stage, and survives every SmoLLM3 post-training stage including the final released model. The bias is regime-dependent, not unidirectional, and it is not introduced by any single training stage.

8 Discussion

Limitations

K and context length are confounded. The total stream length grows as $K \times N$, so increases in K at fixed N co-vary with overall context length. We do not separate the contribution of key count from context length: doing so would require holding $K \times N$ constant and varying the K/N ratio, which we leave for future work.

Ethical Considerations

Acknowledgments

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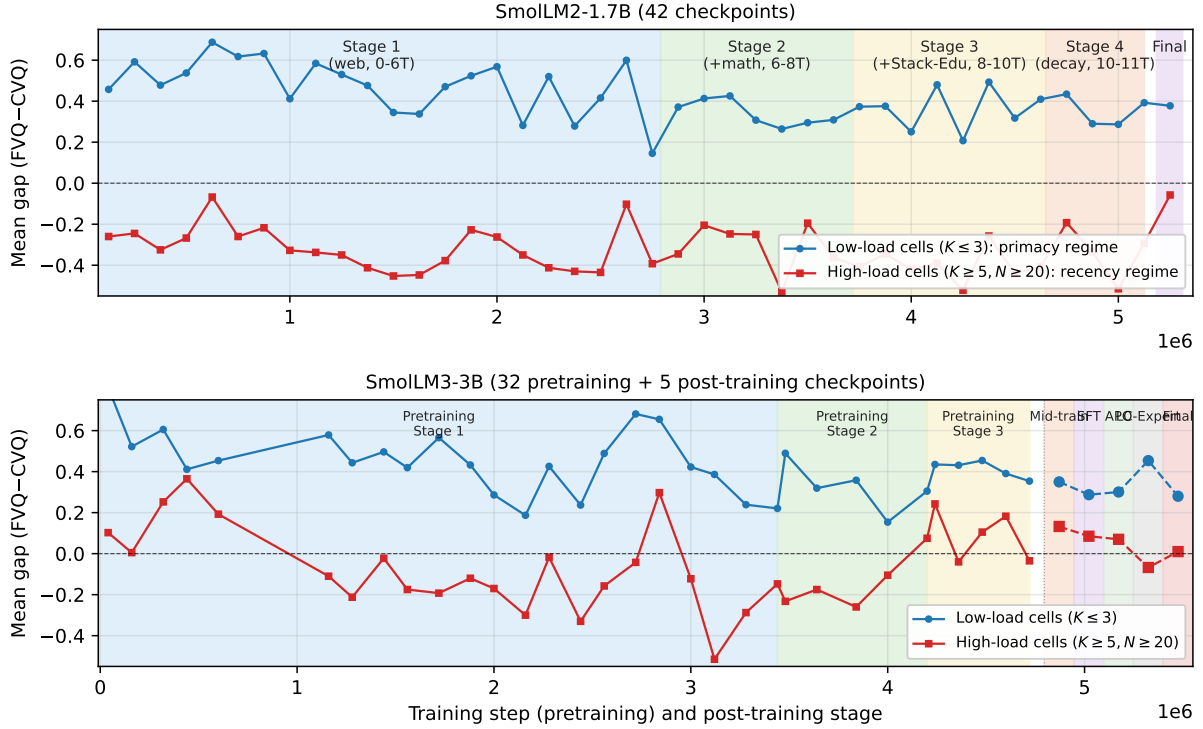


Figure 4: Mean FVQ–CVQ gap across training checkpoints, partitioned by stream-load regime. **Blue (circles)**: mean over low-load cells ($K \leq 3$, 16 cells/checkpoint) — consistently positive across training, the primacy regime. **Red (squares)**: mean over high-load cells ($K \geq 5$, $N \geq 20$, 4 cells/checkpoint) — consistently *negative*, the recency regime in which CVQ exceeds FVQ. **Top**: SmolLM2-1.7B across 41 pretraining steps plus the released checkpoint, with the published four-stage pretraining curriculum (Ben Allal et al., 2025) shown as coloured bands. **Bottom**: SmolLM3-3B across 32 pretraining steps and the four documented post-training stages (mid-training, SFT, APO, LC-expert) plus the released model (Hugging Face Smol Models Research Team, 2025); coloured bands mark each stage. Dashed lines connect post-training stages, which are categorical rather than step-indexed. The regime-dependent bias is present from the earliest available checkpoint and persists through every training stage in both models.

A Dataset Details

We construct three datasets, varying the surface form of the update stream while holding the underlying (K, N) structure fixed. The two key-value variants (ARB and SEM) share schema and grid; the narrative variant (NAR) tests the same task in prose.

Arbitrary-Single (ARB). A pool of 2,300 single-token English words organized into 46 semantic categories (e.g., *color*, *fruit*, *metal*). For each trial, K categories are sampled as keys and N values are drawn without replacement from each category’s pool. Keys and values share no inherent update semantics — there is no reason to expect any particular ordering among values of the same category. This isolates tracking from semantic shortcuts and provides the single-token values required for the mechanistic analysis (§7).

Semantic-Multi (SEM). A pool of 2,403 multi-token values across the same 46 categories, where values are real members of each category (e.g., *gemstone* → *alexandrite*, *amethyst*, ...). Values are multi-token and drawn from a smaller per-category pool (45–70 values), limiting feasible update counts to $N \leq 50$ for most key counts. SEM is the primary behavioural dataset because the realistic key–value semantics rule out trivial template-matching strategies.

Narrative (NAR). Inspired by Kim and Schuster (2023), who embedded entity state updates in programmatic English sequences, we test whether the same tracking task becomes easier when updates appear in natural prose. We use a Dota 2 match commentary dataset in which K game entities (heroes) each have N attribute updates (e.g., gold, kills) woven into match narration; FVQ and CVQ queries ask for the first and current attribute

value of a target hero. (**NOTE: Run GPT-4.1-mini and Claude Haiku on full narrative grid before submission.** Existing results: Qwen2.5-3B-Instruct, Qwen3.5-9B, Gemma-3-4b already done. Claude Haiku has partial results (old runs). Target: Gemma-3-4b, Qwen3.5-9B, Qwen2.5-3B-Instruct, Claude Haiku, GPT-4.1-mini with matched grid for clean cross-format comparison.)

Grid coverage. On the unified grid $K \in \{2, 3, 5, 7, 10, 15, 20, 25, 30\}$, $N \in \{5, 7, 10, 15, 20, 30, 50, 75, 100\}$ (§3), cells where the prompt exceeds a model’s context limit are excluded; cells where the dataset pool is insufficient are also excluded (relevant for SEM at high N). This yields up to 79 feasible cells per model on ARB and up to 63 on SEM.

B Prompt Examples

C Model Details

Model	Family	Params	IT?
Qwen2.5-0.5B-Instruct	Qwen2.5	0.5B	✓
Qwen2.5-1.5B-Instruct	Qwen2.5	1.5B	✓
Qwen2.5-3B-Instruct	Qwen2.5	3B	✓
Qwen2.5-3B	Qwen2.5	3B	×
Qwen3.5-0.8B	Qwen3.5	0.8B	✓
Qwen3.5-2B	Qwen3.5	2B	✓
Qwen3.5-4B	Qwen3.5	4B	✓
Qwen3.5-9B	Qwen3.5	9B	✓
Gemma-3-270m-it	Gemma-3	270M	✓
Gemma-3-1b-it	Gemma-3	1B	✓
Gemma-3-4b-it	Gemma-3	4B	✓
TinyLlama-1.1B	LLaMA	1.1B	✓
StableLM-2-1.6B	StableLM	1.6B	✓
Pythia-410M	Pythia	410M	×
GPT-4.1	OpenAI	—	—
GPT-4.1-mini	OpenAI	—	—
Claude-4.5-Haiku	Anthropic	—	—
Claude-Sonnet	Anthropic	—	—
Gemini-2.5-Flash	Google	—	—
Gemini-2.5-Pro	Google	—	—

Table 2: All 20 models evaluated. IT = instruction-tuned. Parameter counts unavailable for proprietary models.

D Full Model Results

Table 3 reports the FVQ–CVQ gap for all 15 open-weight models at four representative (K, N) cells on Semantic-Multi. Values are $\text{gap} = \text{FVQ} - \text{CVQ}$ with the half-width of an approximate 95% Wilson interval on the difference; up to 200 trials per cell. Empty cells correspond either to the model’s context-length limit (TinyLlama, Mamba) or to dataset pool exhaustion at high N .

Two patterns are visible across the table. First, the gap is positive ($\text{FVQ} > \text{CVQ}$) for the great majority of populated cells across all model families. Second, three models cross into reversal ($\text{gap} < 0$) at large (K, N) : Qwen2.5-3B-Instruct (from $+0.06$ at $K=5, N=30$ to -0.55 at $K=20, N=50$), Qwen2.5-3B base (-0.20 already at $K=5, N=30$), and Qwen3.5-4B (-0.07 at $K=10, N=50$). The Qwen2.5-3B-Instruct reversal is examined mechanistically in §7.

Table 4 reports the same gap statistic for the six proprietary frontier models on the same four (K, N) cells. At small (K, N) all proprietary models hold the FVQ ceiling and exhibit only a small gap; the asymmetry emerges at higher load, where every model shows $\text{gap} > +0.13$ at $(K=20, N=50)$.

E Arbitrary-Single Replication

The Semantic-Multi dataset uses real category members as values, so trivial template matching cannot solve the task; the Arbitrary-Single (ARB) dataset replaces these with single-token words drawn from the same 46 categories, removing semantic coherence between key and value. ARB is also the dataset used for the mechanistic analysis (§7) because single-token values are required for activation patching. Table 5 reports the gap on ARB at the same four (K, N) cells as Table 3.

The qualitative pattern matches: $\text{gap} > 0$ for the great majority of cells, with the same three models (Qwen2.5-3B-Instruct, Qwen2.5-3B base, Qwen3.5-4B) crossing into reversal at large (K, N) . Absolute gap magnitudes are within ± 0.1 of their SEM counterparts in 9 of 12 fully-populated rows, supporting our reliance on ARB for mechanistic analysis: the failure mode visible mechanistically on single-token values is the same failure mode visible behaviourally on realistic multi-token values.

F Format Intervention Data

This appendix reports raw format-intervention data used by §6. We test four prompt formats applied to the same underlying stream content on the Semantic-Multi dataset at $K=10, N=50$ — the hardest cell available in the format sweep ($K \in \{5, 10\}, N \in \{10, 20, 50\}$). Up to 200 trials per (model, format) cell.

Formats. **Plain** (flat_nolabel) is the default condition used elsewhere in the paper: key:

Model	$K=2, N=10$	$K=5, N=30$	$K=10, N=50$	$K=20, N=50$
Qwen2.5-0.5B-Instruct	+0.47 (± 0.11)	+0.59 (± 0.11)	+0.25 (± 0.10)	+0.11 (± 0.09)
Qwen2.5-1.5B-Instruct	+0.56 (± 0.10)	+0.43 (± 0.12)	−0.11 (± 0.13)	+0.04 (± 0.10)
Qwen2.5-3B-Instruct	+0.26 (± 0.12)	+0.06 (± 0.13)	−0.27 (± 0.13)	−0.55 (± 0.11)
Qwen2.5-3B (base)	+0.22 (± 0.08)	−0.20 (± 0.13)	−0.43 (± 0.12)	−0.57 (± 0.10)
Qwen3.5-0.8B	+0.51 (± 0.10)	+0.50 (± 0.12)	+0.45 (± 0.12)	+0.41 (± 0.12)
Qwen3.5-2B	+0.46 (± 0.11)	+0.14 (± 0.13)	−0.01 (± 0.14)	+0.00 (± 0.12)
Qwen3.5-4B	+0.17 (± 0.08)	+0.22 (± 0.11)	−0.07 (± 0.13)	−0.17 (± 0.13)
Qwen3.5-9B	+0.02 (± 0.04)	+0.07 (± 0.10)	+0.10 (± 0.12)	+0.16 (± 0.13)
Gemma-3-270m-it	+0.08 (± 0.06)	+0.26 (± 0.10)	+0.06 (± 0.07)	+0.03 (± 0.04)
Gemma-3-1b-it	+0.71 (± 0.10)	+0.53 (± 0.10)	+0.47 (± 0.10)	+0.53 (± 0.10)
Gemma-3-4b-it	+0.17 (± 0.08)	+0.29 (± 0.10)	+0.41 (± 0.12)	+0.55 (± 0.11)
TinyLlama-1.1B	+0.94 (± 0.05)	−0.04 (± 0.11)	—	—
StableLM-2-1.6B	+0.58 (± 0.10)	+0.92 (± 0.06)	+0.57 (± 0.11)	—
Pythia-410M	+0.29 (± 0.11)	—	—	—

Table 3: Open-weight models on Semantic-Multi at four representative (K, N) cells. Each cell reports gap = FVQ − CVQ with the half-width of an approximate Wilson 95% interval on the difference; up to 200 trials per cell. Empty cells: context-length limit or dataset pool exhaustion.

Model	$K=2, N=10$	$K=5, N=30$	$K=10, N=50$	$K=20, N=50$
GPT-4.1	+0.00 (± 0.06)	+0.05 (± 0.06)	+0.24 (± 0.07)	+0.29 (± 0.07)
GPT-4.1-mini	+0.00 (± 0.05)	+0.04 (± 0.07)	+0.15 (± 0.07)	+0.26 (± 0.07)
Claude-4.5-Haiku	+0.01 (± 0.02)	+0.33 (± 0.07)	+0.48 (± 0.07)	+0.47 (± 0.07)
Claude-Sonnet	+0.00 (± 0.06)	+0.00 (± 0.06)	+0.10 (± 0.07)	+0.26 (± 0.07)
Gemini-2.5-Flash	+0.00 (± 0.05)	+0.08 (± 0.07)	+0.20 (± 0.07)	+0.46 (± 0.07)
Gemini-2.5-Pro	+0.00 (± 0.06)	+0.00 (± 0.06)	+0.00 (± 0.06)	+0.13 (± 0.08)

Table 4: Proprietary frontier models on Semantic-Multi at the same four (K, N) cells as Table 3. Format follows Table 3.

value, one per line, no positional markers. **Labeled** (flat_verbose) tags each entry with an explicit update index: key (update j) value. **Block** groups all K keys per round under a [Update j] header (no per-entry labels). **Landmark** uses the block layout but separates rounds with a − delimiter and no round number, so round boundaries are visually explicit without a numeric anchor.

Gap reduction (Table 6). Both explicit-label formats collapse the FVQ−CVQ gap toward zero for open-weight models: Plain-format gaps of +0.27 to +0.99 become ± 0.10 under Labeled or Block. Proprietary models start with a smaller Plain-format gap (mostly $\leq +0.31$, with Claude-4.5-Haiku at +0.56) but follow the same direction. The Landmark format — which removes numeric labels but keeps round boundaries visible — helps the proprietary models reach near-zero gaps but does not help Gemma-3-4b-it (+0.92) or Qwen3.5 models (+0.52 to +0.73), suggesting that explicit per-update tokens (not just round structure) are what restores CVQ in the open-weight group.

Absolute CVQ accuracy (Table 7). The gap reduction is driven by CVQ rising under labelled formats, not by FVQ falling: under Plain, open-weight CVQ accuracy is ≤ 0.23 for all four reported models; under Block it rises to ≥ 0.89 in every case. Labeled gives a similar lift for three of four open-weight models (Qwen3.5-2B/4B, Qwen2.5-3B-Instruct, and to a lesser extent Gemma-3-4b-it at 0.47). This rules out the alternative explanation that the formats simply confuse the model into uniform random output.

G Training Dynamics Data

This appendix lists the per-checkpoint behavioural numbers behind Figure 4 (§7.1).

Method. For each checkpoint we evaluate on Arbitrary-Single across $K \in \{2, 3, 5, 7\}$, $N \in \{2, 3, 5, 7, 10, 15, 20, 30\}$ (32 cells) with 100 trials per cell. We use the completion prompt format throughout because the chat template returns near-zero accuracy on early instruction-tuned checkpoints (e.g., SmolLM3 it-mid-training under chat_1024 gives RI = PI = 0.001). Per-cell numbers are aggregated to a mean RI, mean PI, and mean gap per checkpoint.

Model	$K=2, N=10$	$K=5, N=30$	$K=10, N=50$	$K=20, N=50$
Qwen2.5-0.5B-Instruct	+0.55 (± 0.11)	+0.46 (± 0.11)	+0.22 (± 0.09)	-0.01 (± 0.06)
Qwen2.5-1.5B-Instruct	+0.61 (± 0.10)	+0.64 (± 0.11)	+0.39 (± 0.11)	+0.06 (± 0.09)
Qwen2.5-3B-Instruct	+0.32 (± 0.12)	+0.18 (± 0.13)	-0.08 (± 0.14)	-0.54 (± 0.11)
Qwen2.5-3B (base)	+0.33 (± 0.09)	-0.10 (± 0.14)	-0.45 (± 0.11)	-0.41 (± 0.11)
Qwen3.5-0.8B	+0.56 (± 0.10)	+0.55 (± 0.11)	+0.26 (± 0.12)	+0.23 (± 0.11)
Qwen3.5-2B	+0.43 (± 0.12)	+0.22 (± 0.13)	+0.02 (± 0.13)	-0.14 (± 0.09)
Qwen3.5-4B	+0.23 (± 0.08)	+0.30 (± 0.12)	-0.22 (± 0.13)	—
Qwen3.5-9B	+0.19 (± 0.08)	+0.21 (± 0.10)	+0.20 (± 0.12)	+0.24 (± 0.13)
Gemma-3-270m-it	+0.02 (± 0.04)	+0.43 (± 0.10)	+0.21 (± 0.09)	-0.01 (± 0.04)
Gemma-3-1b-it	+0.66 (± 0.10)	+0.46 (± 0.10)	—	—
Gemma-3-4b-it	+0.34 (± 0.09)	+0.29 (± 0.09)	+0.61 (± 0.10)	+0.77 (± 0.09)
TinyLlama-1.1B	+0.88 (± 0.07)	+0.25 (± 0.11)	—	—
StableLM-2-1.6B	+0.15 (± 0.08)	—	+0.77 (± 0.09)	—
Pythia-410M	+0.44 (± 0.10)	-0.18 (± 0.08)	—	—

Table 5: Open-weight models on Arbitrary-Single at the same four (K, N) cells as Table 3. Format follows Table 3.

Model	Plain	Labeled	Block	Landmark
<i>Open-weight models</i>				
Qwen2.5-3B-Instruct	+0.27	-0.06	-0.07	+0.45
Qwen3.5-2B	+0.94	-0.07	+0.01	+0.73
Qwen3.5-4B	+0.99	-0.04	+0.00	+0.52
Qwen3.5-9B	—	—	—	—
Gemma-3-4b-it	+0.90	-0.09	+0.10	+0.92
<i>Proprietary models</i>				
GPT-4.1	+0.31	+0.04	+0.02	+0.00
GPT-4.1-mini	+0.23	+0.06	+0.00	-0.22
Claude-4.5-Haiku	+0.56	+0.05	+0.01	+0.01
Claude-Sonnet	+0.20	+0.19	+0.00	-0.04
Gemini-2.5-Flash	+0.24	+0.02	+0.00	+0.05
Gemini-2.5-Pro	+0.00	+0.00	+0.00	-0.45

Table 6: FVQ-CVQ gap by format on Semantic-Multi at $K=10, N=50$. Up to 200 trials per (model, format) cell. A gap near zero indicates the format intervention has equalised first-position and last-position retrieval; large positive gap means the asymmetry persists. Qwen3.5-9B runs incomplete.

SmolLM2-1.7B (Table 8). The 41 pretraining steps span training step 125,000 through 5,125,000; the *final* row is the released SmolLM2-1.7B checkpoint. The gap is positive at every checkpoint; it rises sharply through mid-pretraining (peaking at step 625,000) and partially collapses late in pretraining before stabilising around +0.25639

SmolLM3-3B (Table 9). The pretraining block covers 32 checkpoints across the three documented pretraining stages of SmolLM3 (Hugging Face Smol Models Research Team, 2025) in completion-few-shot format; the post-training block reports the four documented intermediate stages (it-mid-training, it-SFT, it-LC-expert, it-soup-AP0) followed by the released final model. The gap is positive at every checkpoint and at every post-training stage including final.

Prompt-format sensitivity for post-training (Table 10). For SmolLM3 post-training stages we

additionally ran the chat-format prompt. Magnitudes differ between formats but the sign and ordering of the gap are preserved, except for the it-mid-training chat run where the chat template breaks the model entirely (both RI and PI collapse to near zero; the gap is undefined and we omit it from Figure 4).

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Model	Plain	Labeled	Block	Landmark
<i>Open-weight models</i>				
Qwen2.5-3B-Instruct	0.23	0.39	0.92	0.27
Qwen3.5-2B	0.01	0.93	0.98	0.07
Qwen3.5-4B	0.00	0.97	1.00	0.47
Qwen3.5-9B	—	—	—	—
Gemma-3-4b-it	0.01	0.47	0.89	0.00
<i>Proprietary models</i>				
GPT-4.1	0.69	0.96	0.98	0.99
GPT-4.1-mini	0.77	0.94	0.99	1.00
Claude-4.5-Haiku	0.43	0.95	0.99	0.90
Claude-Sonnet	0.80	0.81	1.00	0.99
Gemini-2.5-Flash	0.76	0.98	1.00	0.95
Gemini-2.5-Pro	1.00	1.00	1.00	0.99

Table 7: CVQ (last-position) accuracy by format on Semantic-Multi at $K=10$, $N=50$. Under the Plain format CVQ collapses to near zero for the open-weight models (except Qwen2.5-3B-Instruct, 0.23); explicit-label formats restore CVQ to near-ceiling for most open-weight models and to ceiling for all proprietary models.

Step	RI (FVQ)	PI (CVQ)	Gap	Cells
125,000	0.373	0.201	+0.17	32
250,000	0.502	0.185	+0.32	32
375,000	0.490	0.300	+0.19	32
500,000	0.563	0.254	+0.31	32
625,000	0.577	0.140	+0.44	32
750,000	0.580	0.243	+0.34	32
875,000	0.587	0.220	+0.37	32
1,000,000	0.554	0.366	+0.19	32
1,125,000	0.565	0.318	+0.25	32
1,250,000	0.518	0.327	+0.19	32
1,375,000	0.499	0.355	+0.14	32
1,500,000	0.533	0.449	+0.08	32
1,625,000	0.479	0.410	+0.07	32
1,750,000	0.520	0.307	+0.21	32
1,875,000	0.558	0.255	+0.30	32
2,000,000	0.566	0.233	+0.33	32
2,125,000	0.396	0.347	+0.05	32
2,250,000	0.547	0.342	+0.20	32
2,375,000	0.459	0.442	+0.02	32
2,500,000	0.522	0.426	+0.10	32
2,625,000	0.567	0.193	+0.37	32
2,750,000	0.384	0.472	−0.09	32
2,875,000	0.508	0.367	+0.14	32
3,000,000	0.526	0.331	+0.19	32
3,125,000	0.610	0.362	+0.25	32
3,250,000	0.592	0.440	+0.15	32
3,375,000	0.554	0.509	+0.05	32
3,500,000	0.579	0.442	+0.14	32
3,625,000	0.558	0.481	+0.08	32
3,750,000	0.578	0.461	+0.12	32
3,875,000	0.655	0.446	+0.21	32
4,000,000	0.584	0.511	+0.07	32
4,125,000	0.655	0.395	+0.26	32
4,250,000	0.627	0.590	+0.04	32
4,375,000	0.623	0.340	+0.28	32
4,500,000	0.598	0.467	+0.13	32
4,625,000	0.620	0.412	+0.21	32
4,750,000	0.704	0.427	+0.28	32
4,875,000	0.665	0.505	+0.16	32
5,000,000	0.641	0.552	+0.09	32
5,125,000	0.728	0.469	+0.26	32
final	0.749	0.498	+0.25	32

Table 8: SmolLM2-1.7B per-checkpoint mean RI (FVQ), PI (CVQ), and gap on Arbitrary-Single. 32 cells per checkpoint; 100 trials/cell.

Checkpoint	RI (FVQ)	PI (CVQ)	Gap	Cells
<i>Pretraining (completion_few_shot)</i>				
stage1-step-40,000	0.664	0.074	+0.59	32
stage1-step-160,000	0.743	0.398	+0.34	32
stage1-step-320,000	0.862	0.332	+0.53	32
stage1-step-440,000	0.877	0.429	+0.45	32
stage1-step-600,000	0.813	0.417	+0.40	32
stage1-step-1,160,000	0.757	0.333	+0.42	32
stage1-step-1,280,000	0.720	0.448	+0.27	32
stage1-step-1,440,000	0.833	0.434	+0.40	32
stage1-step-1,560,000	0.786	0.455	+0.33	32
stage1-step-1,720,000	0.721	0.332	+0.39	32
stage1-step-1,880,000	0.622	0.352	+0.27	32
stage1-step-2,000,000	0.686	0.509	+0.18	32
stage1-step-2,160,000	0.614	0.548	+0.07	32
stage1-step-2,280,000	0.716	0.399	+0.32	32
stage1-step-2,440,000	0.656	0.596	+0.06	32
stage1-step-2,560,000	0.757	0.461	+0.30	32
stage1-step-2,720,000	0.793	0.310	+0.48	32
stage1-step-2,840,000	0.845	0.297	+0.55	32
stage1-step-3,000,000	0.794	0.512	+0.28	32
stage1-step-3,120,000	0.769	0.595	+0.17	32
stage1-step-3,280,000	0.681	0.589	+0.09	32
stage1-step-3,440,000	0.697	0.568	+0.13	32
stage2-step-3,480,000	0.750	0.430	+0.32	32
stage2-step-3,640,000	0.755	0.533	+0.22	32
stage2-step-3,840,000	0.748	0.568	+0.18	32
stage2-step-4,000,000	0.818	0.712	+0.11	32
stage2-step-4,200,000	0.797	0.547	+0.25	32
stage3-step-4,240,000	0.831	0.476	+0.35	32
stage3-step-4,360,000	0.815	0.492	+0.32	32
stage3-step-4,480,000	0.797	0.439	+0.36	32
stage3-step-4,600,000	0.824	0.494	+0.33	32
stage3-step-4,720,000	0.804	0.582	+0.22	32
<i>Post-training (completion)</i>				
it-mid-training	0.788	0.534	+0.25	32
it-SFT	0.823	0.628	+0.19	32
it-soup-APO	0.798	0.612	+0.19	32
it-LC-expert	0.827	0.554	+0.27	32
final	0.809	0.645	+0.16	32

Table 9: SmolLM3-3B per-checkpoint mean RI (FVQ), PI (CVQ), and gap on Arbitrary-Single (completion format throughout). 32 cells per checkpoint; 100 trials/cell.

Stage	Completion		Chat (1024)	
	Gap	Cells	Gap	Cells
it-mid-training	+0.25	32	−0.00	32
it-SFT	+0.19	32	+0.29	32
it-soup-APO	+0.19	32	+0.20	32
it-LC-expert	+0.27	32	+0.33	32
final	+0.16	32	+0.18	32

Table 10: SmolLM3-3B post-training stages: gap under completion vs chat_1024 prompt formats. The chat template returns near-zero accuracy at it-mid-training; magnitudes shift at later stages but the gap stays positive under both formats.